

Additional examples of exam questions (including answers)

Question 1

The question

1) The simplest crop growth model originates from measurements showing that for many crops the potential growth rate amounts to $200 \text{ kg ha}^{-1} \text{ d}^{-1}$.

As you know, the Lintul model is based on the assumption that light intensity exponentially decreases with the accumulated LAI in a crop canopy, and that light can directly be related to the increase of biomass through the Light Use Efficiency (LUE).

Show that the Lintul model approximates the value of $200 \text{ kg ha}^{-1} \text{ d}^{-1}$ for a closed canopy, assuming that the light extinction coefficient, k , amounts to 0.6, the $\text{LUE}=3$, and the daily total radiation, DTR, equals $15 \text{ MJ m}^{-2} \text{ d}^{-1}$.

The answer A LAI of $3 \text{ m}^2 \text{ leaf m}^{-2}$ soil surface and a $k = 0.6$ mean that the term $(1.0 - e^{-(k \cdot \text{LAI})})$ in the equation $\text{PARINT} = 0.5 \cdot \text{DTR} \cdot (1.0 - e^{-(k \cdot \text{LAI})})$ amounts to 0.83(47), so PARINT is $0.5 \cdot \text{DTR} \cdot (1.0 - e^{-(k \cdot \text{LAI})}) = 0.5 \cdot 15 \cdot 0.83(47) = 6.26(03) \text{ MJ m}^{-2} \text{ d}^{-1}$, and the growth rate is given by $\text{GTOTAL} = \text{PARINT} \cdot \text{LUE}$, so $6.26(03) \cdot 3 = 18.78 \text{ g m}^{-2} \text{ d}^{-1}$ or $188 \text{ kg ha}^{-1} \text{ d}^{-1}$. This value favourably compares with the rule of thumb of $200 \text{ kg ha}^{-1} \text{ d}^{-1}$. (At a LAI of 4 the growth rate amounts to $205 \text{ kg ha}^{-1} \text{ d}^{-1}$.)

The question

2) In Lintul only one model layer is used to keep track of the soil water for the crop. Explain why this is a reasonable assumption.

The answer This is reasonable, because the roots of the crop integrate the extraction of the water over the profile depth. In reality, dry areas in the soil, where hardly any water is taken up, will be compensated by root water uptake in wetter areas. So, although in reality there are differences of water content with depth, the roots may level out these differences. The model is kept simple due to this integrating effect.

By using one model layer, water profiles can, of course, not be calculated: the water content using one integral will be the same everywhere. {Due to non-linearities in water uptake as a function of WC, there may be differences between models with one layer as compared to models with a number of layers, however.}

Question 2.

The figure below shows a part of the relational diagram of the water balance of the Lintul2 model, except for the names of the state, the flows, their directions and their feedbacks. The circle is an auxiliary (or help) variable to indicate the LAI.

1) Incorporate the names of the state and the rates in the relational diagram.

The answer See drawing. The state is indicated in the rectangle as WA. Furthermore, clockwise, starting on the left part of the figure: interception; rain; irrigation; runoff; evaporation; transpiration; exploration of the water in the soil; drainage.

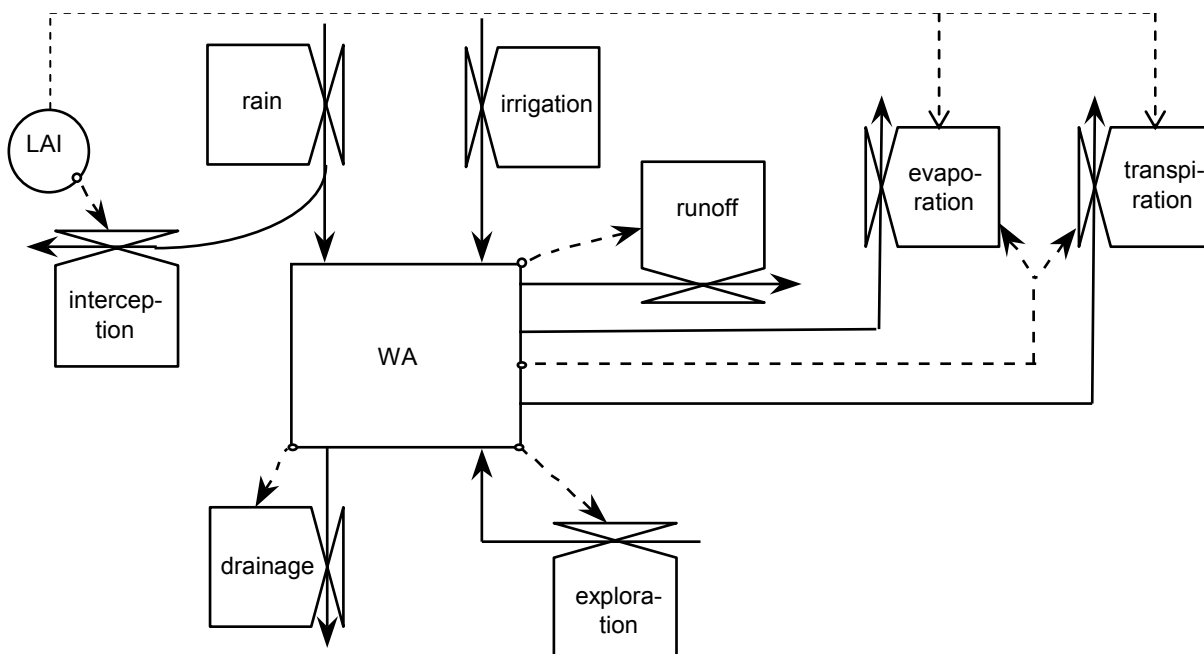
2) Indicate the directions of the flows in the relational diagram.

The answer See the arrows at the end of the full lines in the drawing.

3) Incorporate the feedbacks into the relational diagram. Clearly explain why a feedback occurs. If there is no feedback, also indicate why it is absent. By these explanations you in fact explicitly give the assumptions underlying this model on water dynamics.

Please note that you can only obtain points for this question if you also give explanations.

The answer See drawing and explanations below.



Explanations:

- No feedbacks from the amount of water to the rain and irrigation, because these are the influences from the environment on the soil;

If a student has drawn a feedback from the state variable to the irrigation inflow AND motivates this answer by saying that the farmer may have a water potential measuring instrument to assess if irrigation should and will be applied, I will reckon this as a correct answer.

- Feedback to the runoff flow, because only when the soil is saturated (WA larger than the saturation point) runoff flow starts in the model;
- Feedback to the evaporation, because if the soil dries out evaporation is reduced, for example due to the mulching effect of the top layer of the soil (which may have a thickness of only a few millimeter). Then only water vapour transport occurs, which is by diffusion and very slow;
- Feedback to the transpiration, because if the soil dries out water is not transported to the roots

so fast anymore and crop transpiration is reduced;

- Feedback to the exploration rate, because roots in a dry soil may not grow (in downward direction) anymore and thus no soil is explored;
- Feedback to the drainage, because drainage can only occur when the soil is wet enough, which means above field capacity in this tipping bucket approach of the Lintul-2 model.

- 4) The auxiliary variable (indicated by the circle) is LAI. Indicate which rates are affected by LAI and explain why.

The answer A feedback comes from the LAI from the crop model to (i) the interception of rain, perhaps to the rate of irrigation if the water is given via sprinkler irrigation (but in that case there should also be an interception of that water!), (ii) to the evaporation rate, because that rate is reduced if LAI is higher, and (iii) to the transpiration rate, because of more transpiring leaf surface.

- 5) Explain the essential assumption of the Lintul2 water balance model.

The answer In the Tipping Bucket (TB) approach soil layers are just filled up to their field capacity ($pF=2$, or 100 cm soil water suction). Water in excess of $pF=2$ directly "flows over" to the next compartment, or, if there is only one compartment as in Lintul2 used in the course, water flows out of the soil system considered. So, in fact there is no calculation of flow rates of water due to differences in water potential and gravimetric potential, and the distribution of water over the soil layer(s) is instantaneous. The flow rates are considered as the amount of water flowing to the next layer (or out of the system) divided by the time over which this takes place (1 day).

Which characteristic moisture content data are needed for the tipping bucket model?

The answer Characteristic moisture content data that are needed for the tipping bucket model are: only some of the data of the pF -curve have to be known for the TB-approach (θ_{sat} , $\theta_{at\ pF=2}$, $\theta_{at\ pF=4.2}$).